

TOPIC 1: PROBABILITY BASICS

◆ 1.1 Probability

Definition:

Probability is a numerical measure of the likelihood that an event will occur.

Range:

$$0 \leq P(E) \leq 1$$

- $P(E) = 0 \rightarrow$ Impossible event
 - $P(E) = 1 \rightarrow$ Certain event
-

◆ 1.2 Trial (Experiment)

Definition:

A trial is a process that produces an outcome.

Example:

- Tossing a coin
 - Throwing a dice
-

◆ 1.3 Event

Definition:

An event is the result of a trial.

Example:

- Getting Head in a coin toss
 - Getting 4 on a dice
-

◆ 1.4 Sample Space (S)

Definition:

The sample space is the set of all possible outcomes of a random experiment.

Examples:

- Coin toss:

$$S = \{H, T\}$$

- Dice throw:

$$S = \{1, 2, 3, 4, 5, 6\}$$

◆ 1.5 Exhaustive Events

Definition:

A set of events is said to be exhaustive if at least one of them must occur when the experiment is performed.

Condition:

$$E_1 \cup E_2 \cup E_3 = S$$

Example:

For a dice:

$$E_1 = \{1, 2\}, E_2 = \{3, 4\}, E_3 = \{5, 6\}$$

◆ 1.6 Impossible Event

Definition:

An event that cannot occur in any experiment.

Formula:

$$P(E) = 0$$

Example:

Getting 7 on a dice

◆ 1.7 Mutually Exclusive Events

Definition:

Two events are mutually exclusive if they cannot occur simultaneously.

Condition:

$$A \cap B = \emptyset$$

Example:

In a coin toss:

- Getting Head and Tail together is impossible
-

◆ **1.8 Equally Likely Events**

Definition:

Events that have the same probability of occurrence.

Example:

In a fair coin:

$$P(H) = P(T) = \frac{1}{2}$$

TOPIC 2: PROBABILITY RULES & THEOREMS

◆ **2.1 Complementary Event**

Definition:

The complement of an event A consists of all outcomes that are **not in A** .

Formula:

$$P(A') = 1 - P(A)$$

Example:

If $P(A) = 0.7$, then:

$$P(A') = 0.3$$

◆ **2.2 Addition Theorem**

Definition:

The addition theorem gives the probability that **at least one of the events A or B occurs**.

Formula:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Special Case (Mutually Exclusive Events):

$$P(A \cup B) = P(A) + P(B)$$

👉 Because:

$$P(A \cap B) = 0$$

When to Use:

- Questions with “**OR**”
 - “At least one occurs”
-

♦ **2.3 Multiplication Theorem**

Definition:

The multiplication theorem gives the probability that **both events A and B occur**.

Formula:

$$P(A \cap B) = P(A) \cdot P(B | A)$$

Special Case (Independent Events):

$$P(A \cap B) = P(A) \cdot P(B)$$

When to Use:

- Questions with “**AND**”
 - Sequential events (cards, balls, etc.)
-

♦ **2.4 Conditional Probability**

Definition:

The probability of occurrence of an event A , given that event B has already occurred.

Formula:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, P(B) \neq 0$$

Meaning:

We calculate probability **under a condition**

◆ 2.5 Bayes' Theorem

Definition:

Bayes' theorem is used to find the **reverse probability** (probability of cause given result).

Formula:

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

When to Use:

- Given **different sources (bags, machines, hospitals)**
 - Asked: "Given result, find source"
-

◆ 2.6 Independent Events

Definition:

Two events are independent if the occurrence of one **does not affect** the other.

Condition:

$$P(A \cap B) = P(A) \cdot P(B)$$

Axioms of Probability

Definition:

Axioms of probability are the **basic rules** that are always true for any probability model.

Axiom 1: Non-negativity

$$P(A) \geq 0$$

👉 The probability of any event is always **greater than or equal to zero**.

Axiom 2: Total Probability

$$P(S) = 1$$

👉 The probability of the **sample space (certain event)** is always 1.

Axiom 3: Additivity (for mutually exclusive events)

$$P(A \cup B) = P(A) + P(B)$$

👉 If events A and B **cannot occur together**, then the probability of either occurring is the sum of their probabilities.

🔥 Quick Understanding (VERY IMPORTANT)

Situation	Formula
Opposite event	Complement
OR	Addition
AND	Multiplication
Given condition	Conditional
Reverse probability	Bayes

TOPIC 3: RANDOM VARIABLES

◆ 3.1 Random Variable

Definition:

A random variable is a function that assigns a **real number** to each outcome of a random experiment.

$$X: S \rightarrow \mathbb{R}$$

Example:

Two coin tosses:

Outcome	X (No. of Heads)
HH	2
HT	1
TH	1
TT	0

👉 Here, X is a random variable

♦ 3.2 Types of Random Variables

✅ (a) Discrete Random Variable

Definition:

A random variable that takes **finite or countable values**.

Characteristics:

- Countable values
 - No continuous range
-

Examples:

- Number of heads
 - Number of students
 - Number of defective items
-

♦ 3.3 Probability Mass Function (PMF)

Definition:

A PMF gives the probability of each possible value of a **discrete random variable**.

Notation:

$$P(X = x) = p(x)$$

Conditions:

1. Non-negative:

$$p(x) \geq 0$$

2. Total probability:

$$\sum p(x) = 1$$

Example:

X **0** **1** **2**

P(X) 0.2 0.5 0.3

Check:

$$0.2 + 0.5 + 0.3 = 1 \quad \checkmark$$

♦ 3.4 Continuous Random Variable**Definition:**

A random variable that takes **infinite values in an interval**.

Key Property:

$$P(X = x) = 0$$

Examples:

- Height
 - Weight
 - Temperature
-

◆ 3.5 Probability Density Function (PDF)

Definition:

A function $f(x)$ that describes the probability distribution of a **continuous random variable**.

Conditions:

1. Non-negative:

$$f(x) \geq 0$$

2. Total probability:

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

◆ 3.6 Cumulative Distribution Function (CDF)

Definition:

The CDF gives the probability that X is less than or equal to a value.

Formula:

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$

◆ 3.7 Probability of Interval

Probability that X lies between two values

$$P(a < X < b) = \int_a^b f(x) dx$$

Quick Understanding

Concept	Meaning
Random Variable	Converts outcomes $\rightarrow$ numbers
Discrete	Countable values
Continuous	Infinite range
PMF	Probability for discrete
PDF	Probability for continuous
CDF	Accumulated probability

Important Exam Points

- PMF $\rightarrow$ use Σ (summation)
- PDF $\rightarrow$ use $\int$ (integration)
- Always check:
 - PMF $\rightarrow$ sum = 1
 - PDF $\rightarrow$ integral = 1

TOPIC 4: MATHEMATICAL EXPECTATION & VARIANCE

4.1 Mathematical Expectation (Mean)

Definition:

The expectation (or mean) of a random variable is the **average value** of all possible outcomes, weighted by their probabilities.

For Discrete Random Variable

Formula:

$$E(X) = \sum x \cdot P(x)$$

Example:

X 1 2 3

P(X) 0.2 0.5 0.3

$$E(X) = 1(0.2) + 2(0.5) + 3(0.3)$$

 For Continuous Random Variable

Formula:

$$E(X) = \int_{-\infty}^{\infty} xf(x) dx$$

 4.2 Expectation of a Function

Definition:

The expected value of a function of a random variable.

Formula:

Discrete:

$$E[g(X)] = \sum g(x) \cdot P(x)$$

Continuous:

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x) dx$$

 4.3 Second Moment

Definition:

The expected value of the square of the random variable.

Formula:

$$E(X^2) = \sum x^2 P(x) (\text{discrete})$$
$$E(X^2) = \int x^2 f(x) dx (\text{continuous})$$

◆ 4.4 Variance

Definition:

Variance measures the **spread or dispersion** of values from the mean.

Formula:

$$Var(X) = E(X^2) - [E(X)]^2$$

◆ 4.5 Standard Deviation

Definition:

Square root of variance.

Formula:

$$\sigma = \sqrt{Var(X)}$$

🔥 Important Properties

1. Linearity of Expectation:

$$E(aX + b) = aE(X) + b$$

2. Variance Property:

$$Var(aX + b) = a^2Var(X)$$

🔥 Quick Understanding

Concept	Meaning
Mean $E(X)$	Average value
$E(X^2)$	Square expectation

Variance	Spread
Standard deviation	$\sqrt{\text{variance}}$

⚠ Important Exam Points

- Always calculate $E(X)$ first
- Then find $E(X^2)$
- Then apply variance formula
- Don't forget squaring mean

🎯 Most Common Questions

- Find mean
- Find variance
- Find $E(X^2)$
- Given PMF/PDF $\rightarrow$ calculate

TOPIC 5: MOMENT GENERATING FUNCTION (MGF)

◆ 5.1 Definition

Definition:

The Moment Generating Function (MGF) of a random variable X is defined as:

$$M_X(t) = E(e^{tX})$$

👉 It is used to find **mean, variance, and higher moments**

◆ 5.2 MGF for Discrete Random Variable

Formula:

$$M_X(t) = \sum e^{tx} \cdot P(x)$$

◆ 5.3 MGF for Continuous Random Variable

Formula:

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} f(x) dx$$

◆ 5.4 Expansion (Why it works)

$$e^{tX} = 1 + tX + \frac{t^2 X^2}{2!} + \frac{t^3 X^3}{3!} + \dots$$

Taking expectation:

$$M_X(t) = 1 + tE(X) + \frac{t^2 E(X^2)}{2!} + \dots$$

👉 So it generates moments:

- $E(X)$
 - $E(X^2)$
 - $E(X^3)$
-

◆ 5.5 Mean using MGF

$$E(X) = M'_X(0)$$

👉 First derivative at $t = 0$

◆ 5.6 Second Moment

$$E(X^2) = M''_X(0)$$

◆ 5.7 Variance using MGF

$$\text{Var}(X) = M''_X(0) - [M'_X(0)]^2$$

◆ 5.8 Property of MGF

If:

$$Y = aX + b$$

Then:

$$M_Y(t) = e^{bt} \cdot M_X(at)$$

🔥 Steps to Solve Problems

1. Write:

$$M_X(t) = E(e^{tX})$$

2. Use:

- Summation (discrete)
- Integration (continuous)

3. Simplify expression

4. Differentiate:

- Once $\rightarrow$ mean
- Twice $\rightarrow$ variance

5. Substitute:

$$t = 0$$

Let's prove it with logic

◆ Q1 → Conditional

👉 Words:

“given that first ball is red”

➡ Keyword: “**given that**”

👉 That directly means:

$$P(A | B)$$

So → Conditional Probability

◆ Q2, Q4, Q5 → Bayes

👉 Words:

“it is found to be red, find from which bag/machine”

➡ You only know **result**, not source

👉 That is:

Result → Find Cause

➡ This is exactly **Bayes**

◆ Q3, Q8 → Binomial

👉 Words:

- “fires 6 times”
- “number of heads in 4 tosses”
- “at most / at least / exactly”

➡ Repeated trials + success/failure

👉 That is **Binomial Distribution**

◆ Q6, Q7 → Discrete Random Variable

👉 Words:

- “minimum of two dice”

- “maximum of two dice”
- “find distribution, mean, variance”

→ You must:

- list values
- find probabilities

👉 That is **PMF (discrete)**

♦ **Q9 → PMF + CDF**

👉 Words:

- “given table”
- “find k”
- “ $P(X \leq x)$ ”

→ Table → PMF

→ $P(X \leq x)$ → CDF

♦ **Q10–Q15 → Continuous (PDF)**

👉 Words:

- “ $f(x) = \dots$ ”
- “check if PDF”
- “find mean, variance”
- “integral”

→ Integration involved

👉 That is **Continuous Random Variable**

🔥 **Core Logic (VERY IMPORTANT)**

👉 You don’t memorize topics

👉 You identify **keywords**

💡 **Master Trick**

Keyword	Topic
given that	Conditional
find source	Bayes
at most / at least	Binomial
dice min/max	Discrete
table (P(x))	PMF
f(x), integral	PDF